

ASSIGNMENT ON CSE-423

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Section : 18

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Answer to the Question No.1

S

(a)

(Light source)

S (6, -4, 10), Intensity = 40, $r = 60$

View (0, 0, 5)

For ambient method we account, $I_{\text{aa}} = I_{\text{a}} k_{\text{a}}$ and doesn't depend on light or view direction.

The specular contributes to $I_{\text{s}} = I k_{\text{s}} ((R \cdot V))^2$

where L is the Light direction and

R is the mirror reflection of L about the normal and V is the unit vector pointing to the camera. The diffuse model is

$I_{\text{d}} = k_{\text{d}} I (\hat{L} \cdot N, 0)$ where L is the light source and N , the normal.

The angle between R and V is α . As we move our attention from the exact reflection direction, α grows so it tends to be zero. When $R \cdot V < 0$, the specular term proceeds to be 0, so no highlight is seen.

(b)

Plane $\rightarrow z=2$

$$N = (0, 0, 1)$$

So, the unit normal vector at the point of reflection is $(0, 0, 1)$ or

$$\frac{1\hat{k}}{\sqrt{1}}$$

(c)

We know,

$$L + R = 2(\hat{L} \cdot \hat{n}) \cdot \hat{n}$$

$$L = (S - P) = (6-1, -4-3, 10-2) = (5, -7, 8)$$

$$\text{Unit Vector} = \frac{5\hat{i} + (-7)\hat{j} + 8\hat{k}}{\sqrt{138}}$$

$$\text{So, } \hat{L} \cdot \hat{n} = \frac{-8}{\sqrt{138}}$$

$$\text{Thus, } R = 2\left(\frac{-8}{\sqrt{138}}\right)(0, 0, 1) - \frac{5\hat{i} - 7\hat{j} + 8\hat{k}}{\sqrt{138}}$$

$$= \frac{-16\hat{k}}{\sqrt{138}} - \left(\frac{5\hat{i} - 7\hat{j} + 8\hat{k}}{\sqrt{138}}\right)$$

$$= -\frac{5\hat{i}}{\sqrt{138}} - \frac{7\hat{j}}{\sqrt{138}} - \frac{4\sqrt{138}}{23}\hat{k}$$

(d)

The attenuation factor at the point (1,3,2) due to the point light source is

$$d = \sqrt{138}$$

$$f_{att} = 1 - \left(\frac{\sqrt{138}}{60} \right)^2$$
$$= \frac{577}{600} \Rightarrow 0.961$$

(e)

Total Phong Intensity at P is required to be found

Ambient

$$I_a = 0.3 \times 8 = 2.4$$

Diffuse

$$\hat{L} \cdot \hat{n} = \frac{8}{\sqrt{138}} = 0.68 \rightarrow 0.6 \times 40 \times 0.68$$
$$= 16.32$$

$$V \rightarrow (-1, -3, 3)$$

$$\hat{V} = \frac{-1\hat{i} - 3\hat{j} + 3\hat{k}}{\sqrt{19}}$$

Specular

$$(R \cdot V)^6 = 0.00000923$$

$$R_0, I_s = 0.5 \times 40 \times 0.00000923 \\ = 0.0019$$

Now we apply attenuation factor

$$I_{att} = (I_d + I_s) = 0.9617 (16.35 + 0.0019) \\ = 15.73$$

So, Total intensity

$$I_{total} = 2.4 + 15.73 \\ = 18.1$$

(5)

For, halfway vector (H) method.

$$(R \cdot V)^n \rightarrow \text{we get specular } 0.0019$$

We know,

$$H = \frac{L + V}{\|L + V\|} \rightarrow 0.7259 \rightarrow (N \cdot H)$$

$$S_0, (N \cdot H)^5 = 0.2015$$

$$I_{II} \text{ with attenuation factor} = 0.5 \times 10 \times 0.2015 \times 0.06 = 4.03$$

So, no the half way method gives us 4.03 while Phong reflection method gave us 0.002.

Answer to the Question No.2

(a)

The First curve portion has 4 control points. and 2nd portion also has 4 control

$$S \rightarrow P_1, P_2, P_3, P_4$$

$$V \rightarrow P_5, P_4, P_6, P_7$$

For continuity C1 at joining point P5

$$At \quad t=1,$$

$$S'(C1) = 3(P_5 - P_3)$$

At $t=0$ for V ,

$$V'(0) = 3(P_4 - P_5)$$

$$\text{Setting } S'(1) = V'(0)$$

$$P_4 = 2P_5 - P_3$$

So, we get $P_4 = 2P_5 - P_3$

(b)

So, the first Bezier segment passes through a specific point $Q = (3.5, 1.5)$

When $t=0.5$

$$S(t) = (1-t)^3 P_1 + 3t(1-t)^2 P_2 + 3t^2(1-t) P_3 + t^3 P_5$$

$$(3.5, 1.5) = P_1(0.125) + P_2(0.375) + P_3(0.125) + P_5(0.125)$$

$$= P_1(0.125) + P_2(x_2, y_2) + P_3(0.375) + P_5(0.125)$$

So, in terms of x ,

$$(3 \cdot 5) = (0.125 \times P_1 x) + P_2 (x_2) + 10 \cdot 375 \times P_3 x_3 + P_5 (0.125 \times P_5 x_5)$$

In terms of y ,

$$(1 \cdot 5) = (0.125 \times P_1 y_1) + P_2 (y_2) + (0.375 \times P_3 y_3) + P_5 (0.125 \times P_5 y_5)$$

So,

$$P_2 = \left(\frac{8Q - P_1 - 3P_3 - P_5}{3} \right)$$

$$P_2 = \left(\frac{8 \times 3 \cdot 5 - x_1 - 3x_3 - x_5}{3}, \frac{8 \times 1 \cdot 5 - y_1 - 3y_3 - y_5}{3} \right)$$

(c)

So, for C^2 continuity at P_5

$$S''(1) = 6 (P_5 - 2P_3 + P_2)$$

$$V''(0) = 6 (P_6 - 2P_4 + P_5)$$

So, we know, $S''(1) = V''(0)$

To satisfy C^2 continuity we should make sure C^1 also is fulfilled which is

$$P_4 = 2P_5 - P_3$$

So,

$$P_6 = P_2 - 2P_3 + 2P_4$$

So, we get

$$P_4 = 2P_5 - P_3$$

$$P_6 = P_2 - 2P_3 + 2P_4$$

So, the C^1 and C^2 derivative all join..